

GrowMe - My Activity Tracker

Code

GitHub: <https://github.com/hyewony-dev/growme>

Abstract

GrowMe is a personal activity tracking application that helps users record daily activities and track their personal growth over time. Users can create or choose activities such as going to the gym, reading, attending bootcamp, showering, or completing productive tasks. Each activity has a point value and belongs to a category such as fitness, intellect, hygiene, productivity, or mental well-being. When a user completes an activity, the system creates an activity log and adds the activity points to the user's total score. Based on the total score, the user can level up. The goal of the project is to turn daily habits into a simple gamified self-growth system.

It calculates total points and levels. In the future, it may also calculate category-based statistics.

Requirements

Core Requirements:

- A user can register themselves with a username, email and password.
- A user can see available activities.
- A user can log a completed activity.
- When a user logs an activity, activity points are added to the user's total points.
- The user's level is calculated based on total points divided by 100.
- A user can see their activity log history.
- A user can see their total points and current level.
- A user can delete a logged activity.

Future Features:

- A user can delete their account, and all their activity logs are deleted as well.
- A user can create custom activities.
- A user can edit or delete custom activities.
- A user can set the category and point value of each activity.
- A user can see statistics by category.

- A user can view their growth through a radar chart.
- A user can earn badges or achievements.

API

POST /users		
Data IN: { username, email, password }	Data out: 200 "User was created"	Data out: 403: "User already exists"

GET /activities			
Authorization: token			
Data IN: empty	Data out: [{ id, name, description, categoryId, points }]	Data out: 403: "Token is wrong."	Data out: 404: "No activities found."

POST /activity-logs			
Authorization: token			
Data IN:	Data out:	Data out:	Data out:

{ activityId, note }	200: { "Activity was logged. Points were added." }	403: "Token is wrong."	404: "Activity does not exist."
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GET /users/{userId}/activity-logs			
Authorization: token			
Data IN: empty	Data out: 200: { logs: [{ activityName, earnedPoints, loggedAt }] }	Data out: 200: { logs: [] }	Data out: 403: "The token is wrong."

GET /users/{userId}/progress		
Authorization: token		
Data IN: empty	Data out: 200: { username, totalPoints, level }	Data out: 403: "The token is wrong."

PATCH /activity-logs/{id}			
Authorization: token			
Data IN:	Data out: 200:	Data out: 403: "Token is wrong."	Data OUT:

{ note }	"Activity log was updated"		404: "Activity log does not exist"
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DELETE /activity-logs/{id}			
Authorization: token			
Data IN: empty	Data out: 200: "Activity log was deleted. Points were subtracted."	Data out: 403: "Token is wrong or this log does not belong to this user."	Data out: 404: "Activity log does not exist."

Database Design

User

- id
- username
- email
- passwordHash
- totalPoints
- level

ActivityCategory

- id
- name

Activity

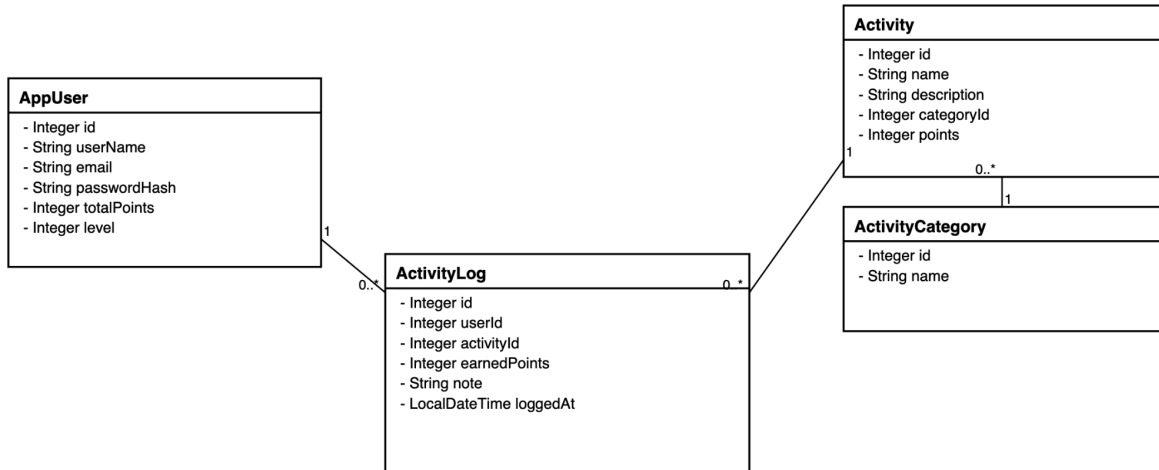
- id
- name
- description
- categoryId
- points

ActivityLog

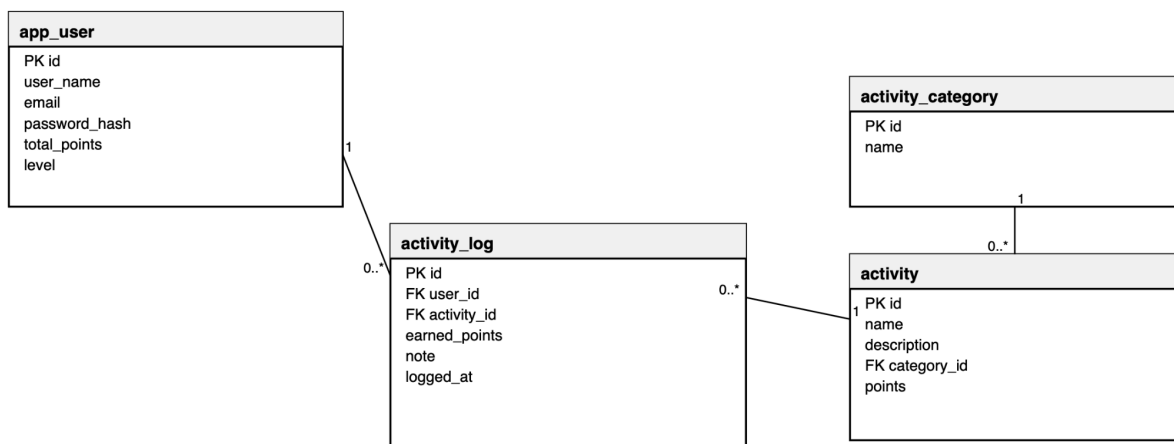
- id
- userId
- activityId
- earnedPoints

- note
- loggedAt

GrowMe - Class Diagram



GrowMe - ERD



GrowMe - Use Case Diagram

